

Advanced Optimization

Chapter 11 — Nonlinear Equations

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Reading: Nocedal & Wright, *Numerical Optimization*, 2nd ed., Chapter 11
Companion to Ch. 10 — taught side by side.

Square systems: $r(x) = 0$, and how this differs from optimization

$r : \mathbb{R}^n \rightarrow \mathbb{R}^n$ continuously differentiable, find x^* with $r(x^*) = 0$.

Example 11.1 (aircraft equilibrium). $F(x) = Ax + \phi(x) = 0$: 5 force-balance equations, 8 unknowns; fixing the 3 controls leaves a square 5×5 system per setting. (Same shape as a power flow: fixed injections, solve for the state.)

Differences from optimization

- ▶ Quadratic convergence needs only *first* derivatives (J), not second.
- ▶ All roots are mathematically **equally good** — no “lower objective” to prefer one; if the root found is physically wrong, fix the *model*.
- ▶ No natural merit function — one must be manufactured (§11.2), and it has traps.
- ▶ Quasi-Newton methods: useful, but less central than in Ch. 6.

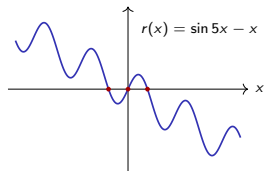


Figure 11.1: three roots, none “better”.

Plan: local algorithms (§11.1: Newton, inexact, Broyden, tensor) → globalization (§11.2: merit functions, line search, trust region) → continuation (§11.3).

Newton's method: linearize the residual (Algorithm 11.1)

Theorem 11.1 (Taylor, integral form)

$$r(x + p) = r(x) + \int_0^1 J(x + tp) p \, dt \quad (11.4)$$

Approximate the integral by $J(x_k)p$: linear model $M_k(p) = r_k + J_k p$ (11.5); set it to zero:

$$\textbf{Algorithm 11.1:} \quad J(x_k) p_k = -r(x_k), \quad x_{k+1} = x_k + p_k. \quad (11.6)$$

A *linear* model, not quadratic — it normally has a solution, and it is enough for fast convergence.

This one algorithm has been chasing us all semester

Apply it to $r = \nabla f$: Newton for unconstrained optimization (2.15). Apply it to KKT systems: SQP (Ch. 18) and interior-point steps (Ch. 19). *Newton power flow is Algorithm 11.1 verbatim.*

Shortcomings (each gets fixed later)

Erratic from remote starts (§11.2); J_k singular $\Rightarrow$ step undefined (11.44/TR); J hard to get (Broyden, Ch. 8 AD); exact solves too costly (inexact Newton); **degenerate roots**: $r(x) = x^2$ gives $x_k = x_0/2^k$ — merely *linear*, rate $\frac{1}{2}$.

Local rates (Theorem 11.2), and the inexact version (Theorem 11.3)

Theorem 11.2

x^* a *nondegenerate* root ($J(x^*)$ nonsingular), x_k near x^* : $x_{k+1} - x^* = o(\|x_k - x^*\|)$ (Q-superlinear); with J Lipschitz: $x_{k+1} - x^* = O(\|x_k - x^*\|^2)$ (Q-quadratic).

Proof shape: $r_k = J_k(x_k - x^*) + w$ with $w = o(\|x_k - x^*\|)$ (11.10–11.12); multiply by J_k^{-1} (bounded near x^*) and the Newton step cancels the linear term exactly. Lipschitz upgrades $o(\cdot)$ to $O(\|\cdot\|^2)$.

Inexact Newton (Framework 11.2 = Ch. 7's dial, verbatim)

Accept any p_k with $\|r_k + J_k p_k\| \leq \eta_k \|r_k\|$ (11.17). Theorem 11.3: η small $\Rightarrow$ Q-linear; $\eta_k \rightarrow 0 \Rightarrow$ Q-superlinear; $\eta_k = O(\|r_k\|) + \text{Lipschitz} \Rightarrow$ Q-quadratic.

- ▶ Inner solver: **GMRES or another Krylov method** — CG does *not* apply (J is square but not symmetric positive definite). Restarts keep memory bounded.
- ▶ Products Jd : finite difference (8.11) at one residual evaluation, or forward-mode AD exactly (Ch. 8). Nothing needs the matrix J itself.

Broyden's method: the secant idea without symmetry

From (11.4): $y_k = r_{k+1} - r_k = \int_0^1 J(x_k + ts_k)s_k dt \approx J(x_{k+1})s_k$ (11.26) — so demand of the approximation B_{k+1} the **secant equation**

$$B_{k+1}s_k = y_k \quad (11.27)$$

(n equations, n^2 unknowns — underdetermined for $n > 1$; the $n = 1$ case is the classical scalar secant method).

Broyden update, and Lemma 11.4

$$B_{k+1} = B_k + \frac{(y_k - B_k s_k)s_k^T}{s_k^T s_k} \quad (11.28)$$

is the *least-change* secant update: among all B with $Bs_k = y_k$, it minimizes $\|B - B_k\|_2$. Proof in two lines:

$$\|B_{k+1} - B_k\| = \left\| (B - B_k) \frac{s_k s_k^T}{s_k^T s_k} \right\| \leq \|B - B_k\| \text{ since } \|ss^T/s^T s\| = 1.$$

Same principle as Ch. 6, different constraints

BFGS: least change subject to secant + *symmetry* + positive definiteness (rank two needed). Broyden: J is not symmetric, so nothing to preserve — plain Frobenius least change, and **rank one suffices**. One idea, two geometries.

Broyden in practice (Algorithm 11.3, Theorem 11.5)

Algorithm 11.3

$B_k p_k = -r_k$; line search α_k along p_k ; $s_k = x_{k+1} - x_k$, $y_k = r_{k+1} - r_k$; update by (11.28).

Theorem 11.5

Under Theorem 11.2's assumptions, if $\|x_0 - x^*\| \leq \delta$ and $\|B_0 - J(x^*)\| \leq \epsilon$, Broyden converges Q-superlinearly.

- ▶ The B_0 condition has teeth: unlike BFGS ($H_0 = I$ is fine), Broyden needs a good start — use $J(x_0)$ or a finite-difference approximation of it.
- ▶ B_k goes dense even for sparse J ; large $n \Rightarrow$ limited-memory form via Sherman–Morrison–Woodbury (Ch. 7 ideas).

Convergence of $\|r(x_k)\|$ on (11.31):

k	Broyden	Newton
0	7.4	7.4
1	0.59	0.59
2	2.0×10^{-2}	2.3×10^{-3}
3	2.1×10^{-3}	1.6×10^{-6}
4	3.7×10^{-4}	2.2×10^{-15}
6	4.9×10^{-8}	
8	1.1×10^{-18}	

Twice the iterations, no Jacobian evaluations — the standard trade.

Tensor methods (keyword). Augment the linear model with a low-rank quadratic term $\frac{1}{2} T_k p p$ chosen to *interpolate recent residuals* (11.34) — little storage, no second derivatives, and markedly better on (near-)degenerate roots ($\text{rank } J(x^*) = n-1, n-2$).

Merit functions: manufactured, and booby-trapped

Pure Newton/Broyden from remote starts: divergence, blow-up, even **cycling**

($r(x) = -x^5 + x^3 + 4x$: from $x_0 = 1$, Newton oscillates $1 \rightarrow -1 \rightarrow 1 \rightarrow \dots$ forever, with five perfectly good roots nearby).

To globalize we need a scalar measure of progress. The standard choice:

$$f(x) = \frac{1}{2} \|r(x)\|^2 \quad (11.35)$$

— every root is a global minimizer with $f = 0$. But not every local minimizer is a root.

The trap, precisely

$\nabla f(x^*) = J(x^*)^T r(x^*) = 0$ with $r(x^*) \neq 0$ forces $J(x^*)$ singular (11.36). So merit-function dead ends live exactly where the Jacobian degenerates — and nonlinear systems are full of singular points.

ℓ_1 merit $\|r\|_1$: nonsmooth; the workhorse of constrained optimization (Ch. 17–18).

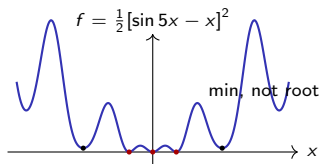


Figure 11.2: roots (red) and impostors (black).

Line search on the merit function

Newton direction is a descent direction for $f = \frac{1}{2}\|r\|^2$ whenever $r_k \neq 0$:

$$p_k^T \nabla f_k = p_k^T J_k^T r_k = -\|r_k\|^2 < 0. \quad (11.38)$$

Wolfe steps give (Theorem 11.6, via Zoutendijk) $\sum_k \cos^2 \theta_k \|J_k^T r_k\|^2 < \infty$; if additionally $\cos \theta_k \geq \delta > 0$ (11.41), then $J_k^T r_k \rightarrow 0$ — and bounded $\|J_k^{-1}\|$ upgrades that to $r_k \rightarrow 0$.

How big is $\cos \theta_k$? Exactly as big as the conditioning allows

$$\text{exact Newton: } \cos \theta_k \geq \frac{1}{\kappa(J_k)} \quad (11.42); \quad \text{inexact } (\eta_k \leq \eta): \cos \theta_k \geq \frac{1 - \eta}{2\kappa(J_k)}.$$

Bounded $\kappa(J_k)$: everything works. Near-singular J_k : **no floor** — and that is not a technicality, as the next slide shows.

Powell's warning, and the repair

Example 11.2 (Powell). The failure is real, not hypothetical

$r(x) = (x_1, 10x_1/(x_1 + 0.1) + 2x_2^2)^T$, unique root 0. Newton with exact line searches from $(3, 1)$ converges to $(1.8016, 0)$ — **not a root, not even a stationary point of f** . Near the x_1 -axis J degenerates, the Newton direction turns $\perp \nabla f$, $\cos \theta_k \rightarrow 0$, and the line search stalls at a point where $-\nabla f$ would still decrease both residuals.

The repair (Algorithm 11.4): regularize when needed

Detect near-singularity / small $\cos \theta_k$ and replace the step by

$$p_k = -(J_k^T J_k + \lambda_k I)^{-1} J_k^T r_k \quad (11.44)$$

with λ_k bounded away from 0 — the $\cos \theta$ floor returns; try $\alpha = 1$ first, Wolfe otherwise. Computed via the augmented QR (10.36), never forming $J^T J$.

But *choosing* λ_k well is exactly the trust region's job — next slide.

Trust region for equations (Algorithm 11.5): Levenberg–Marquardt, fourth appearance

Model = linearized-residual least squares, as in Ch. 10:

$$\min_p m_k(p) = \frac{1}{2} \|r_k + J_k p\|^2 \quad \text{s.t. } \|p\| \leq \Delta_k, \quad (11.45\text{--}11.46)$$

ρ_k from actual vs. predicted decrease in $\|r\|^2$ (11.47); the radius logic is Algorithm 4.1's.

- ▶ $B_k = J_k^T J_k \succeq 0$ always $\Rightarrow$ **no indefinite-model machinery needed**: the dogleg (with Cauchy point (11.48–11.49) and, when J_k is nonsingular, the Newton point $p^J = -J_k^{-1} r_k$) is well-defined, and (11.44) with the right λ is the exact subproblem solution (Lemma 10.2).
- ▶ **Theorem 11.7** (global, via Thms 4.5/4.6): Cauchy-fraction steps give $\liminf \|J_k^T r_k\| = 0$ ($\eta = 0$), and $\rightarrow 0$ for $\eta \in (0, \frac{1}{4})$. Stationarity of the *merit function* — the singular- J caveat never disappears.
- ▶ **Theorem 11.8** (local): if $x_k \rightarrow x^*$ nondegenerate and subproblems are solved exactly, Δ_k stays bounded below, the bound goes inactive, steps become pure Newton — **quadratic convergence**. Globalization does not tax the endgame.

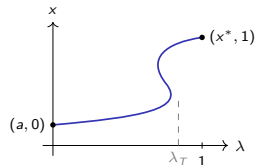
Trust region > line search here, decisively: Example 11.2 defeats exact line searches, while the TR step bends toward $-\nabla f$ automatically as Δ_k shrinks. The solvers agree: MINPACK's `hybrd` (Powell's hybrid = dogleg on (11.46)) is what sits behind `scipy.optimize.root`.

Continuation / homotopy: when Newton's basin is nowhere near you

All Newton-type methods can be trapped by the merit function; continuation methods sidestep the merit function entirely. Deform an easy problem into the real one:

$$H(x, \lambda) = \lambda r(x) + (1 - \lambda)(x - a), \quad H(\cdot, 0) = x - a \text{ (trivial)}, \quad H(\cdot, 1) = r(x). \quad (11.61)$$

- ▶ Naive plan — march $\lambda : 0 \rightarrow 1$, re-solving each time — fails at **turning points** λ_T , where the *zero path* $\{(x, \lambda) : H = 0\}$ bends back (Figure 11.4).
- ▶ Practical fix: parametrize by **arc length** s and follow the path itself, letting λ fall when it must. Tangent $(\dot{x}, \dot{\lambda})$ spans the null space of the $n \times (n+1)$ matrix $[\partial H / \partial x \quad \partial H / \partial \lambda]$ (11.62–11.63); predictor–corrector steps track the curve.



A zero path with a turning point:
 λ must decrease to get through.

- ▶ Verdict: far more work per problem than Newton, but succeeds on problems that defeat merit-function methods; theory guarantees (for a.e. a) a smooth path under mild conditions.
- ▶ The power engineer's version is *continuation power flow*: trace the solution as loading grows, through the nose of the PV curve — a turning point in the flesh.

Ch. 10 $\leftrightarrow$ Ch. 11: one structure, two chapters

	Least squares (Ch. 10)	Nonlinear equations (Ch. 11)
Problem	$\min \frac{1}{2} \ r\ ^2, m \geq n$	$r(x) = 0, m = n$
Linear model of r	GN subproblem (10.26)	Newton equations (11.6)
Exact-model step	$(J^T J)p = -J^T r$	$Jp = -r$
Regularized / TR	LM (10.34)	(11.44) / Algorithm 11.5
Secant method	DGW structured S_k	Broyden (11.28)
Symmetry in update	yes (S_k sym.)	none needed
Merit function	the objective itself	manufactured: $\frac{1}{2} \ r\ ^2$
Its failure mode	large residuals (rate)	local min with J singular (trap)
Local rate	(10.30): residual-dependent	quadratic at nondegenerate roots
Escape hatch	hybrids / Newton	continuation (§11.3)

Two ideas to carry forward

Zero-residual least squares and square nonlinear systems are the same problem; every Ch. 10 algorithm reappears here with $m = n$ and no leftover residual to poison the Hessian. The merit function is a tool, not the truth: its minimizers include impostors, and they live exactly at singular Jacobians.

Homework

Derivations

- 1 Verify the cycling claim: for $r(x) = -x^5 + x^3 + 4x$ and $x_0 = 1$, Newton's iteration oscillates between 1 and -1 .
- 2 Show that $r(x) = x^2$ from $x_0 \neq 0$ gives $x_k = x_0/2^k$, and reconcile with Theorem 11.2 (which assumption fails?).
- 3 Prove Lemma 11.4 (least-change property of the Broyden update), including $\|ss^T/s^T s\|_2 = 1$.
- 4 Derive both $\cos \theta_k$ bounds: (11.42) for exact Newton and $(1 - \eta)/(2\kappa)$ for inexact steps.
- 5 For Example 11.2, compute $J(x)$, verify singularity on the x_1 -axis, and show $(1.8016, 0)$ is not stationary for $f = \frac{1}{2}\|r\|^2$.

Implementation

6. Solve the aircraft system of Example 11.1 (controls fixed at your choice of values) by Newton with backtracking on $\frac{1}{2}\|r\|^2$, and by Broyden with $B_0 = J(x_0)$. Reproduce a Table-11.1-style comparison of $\|r_k\|$: quadratic vs. superlinear tails.
7. Reproduce Powell's failure: exact-line-search Newton on Example 11.2 from $(3, 1)$. Then fix it twice — once with (11.44) and adaptive λ_k , once with the dogleg trust region (Algorithm 11.5) — and compare trajectories.
8. *Optional*: Algorithm 11.5 on a 3-bus power flow ($r(\theta, V) = 0$ mismatch form); compare plain Newton from a flat start vs. a stressed loading — which safeguard fires?

Next: Ch. 12 — *constrained optimization begins: the theory of what optimality even means.*